

Figure 3.17 Automatic quantizing linked with the ColorBrewer 3-red scale produces distinct visual categories in the red family.

```
function buttonClick(datapoint) {
    var maxValue = d3.max(incomingData, function(el) {
            return parseFloat(el[datapoint ]);
        });
    var colorQuantize = d3.scale.quantize()
            .domain([0,maxValue]).range(colorbrewer.Reds[3]);
    var radiusScale = d3.scale.linear()
            .domain([0,maxValue]).range([2,20]);
        d3.selectAll("g.overallG").select("circle").transition()
            .duration(1000).style("fill", function(p) {
            return colorQuantize(p[datapoint ]);
        }).attr("r", function(p) {
            return radiusScale(p[datapoint ]);
        });
};
```

Color is important, and it can behave strangely on the web. Colorblindness, for instance, is a key accessibility issue that most of the colorbrewer scales address. But even though color use and deployment is complex, smart people have been thinking about color for a while, and D3 takes advantage of that.

3.3 Pregenerated content

It's neither fun nor smart to create all your HTML elements using D3 syntax with nested selections and appending. More importantly, there's an entire ecosystem of tools out there for creating HTML, SVG, and static images that you'd be foolish to ignore just because you're using D3 for your general DOM manipulation and information visualization. Fortunately, it's straightforward and easy to load externally generated resources—like images, HTML fragments, and pregenerated SVG—and tie them into your graphical elements.

3.3.1 Images

In chapter 1, I noted that GIFs, despite their resurgent popularity, aren't useful for a rich interactive site. But that doesn't mean you should get rid of images entirely. You'll find that adding images to your data visualizations can vastly improve them. In SVG, the image element is `<image>`, and its source is defined using the `xlink:href` attribute if it's located in your directory structure.

We have files in our images directory that are PNGs of the respective flags of each national team. To add them to our data visualization, select the `<g>` elements that have the team data already bound to them, and add an SVG image: